Lab commands — test Lifecycle Manager from the start

Host: dh2 `172.22.1.2` (BlueField-2, k3s `v1.36.4+k3s1`) **Date:** 2026-10-25 **dh2 has no** `go` **on PATH.** Compile on the Mac, run the binary on dh2. Job files only in `/tmp/lab-job-lcm-from-start/`. Delete after. Do **not** helm-uninstall DPF, do **not** run mlxfwreset, do **not** apply AMD stubs, do **not** copy `opi_lab_key`.

0. Hop onto the lab

This Mac is the VPN Mac (F5 Connected)

```
scutil --nc start "F5 Access"
ping -c 3 172.22.1.2

SSH='ssh -o IdentityAgent=none -o IdentitiesOnly=yes \
  -o IdentityFile=$HOME/.ssh/opi_lab_key -o UseKeychain=yes \
  -o KexAlgorithms=curve25519-sha256'

$SSH root@172.22.1.2 'hostname && nproc && free -h && k3s --version'
```

Other Mac (not on VPN) — hop through the VPN Mac

```
ssh kartikey0104@Kartikeys-MacBook-Pro.local \
  'ssh -o IdentityAgent=none -o IdentitiesOnly=yes \
     -o IdentityFile=$HOME/.ssh/opi_lab_key -o UseKeychain=yes \
     -o KexAlgorithms=curve25519-sha256 \
     root@172.22.1.2 "hostname && k3s --version"'
```

1. On the Mac — compile (lab cannot `go test`)

```
cd /Users/kartikey0104/Desktop/opi-dpf-adapter/opi-nvidia-dpf-adapter
export GOTOOLCHAIN=go1.26.10

go test ./pkg/lifecycle/ -count=1 -timeout 4m
go test -race ./pkg/lifecycle/ -count=1 -timeout 4m
make govulncheck

CGO_ENABLED=0 GOOS=linux GOARCH=amd64 GOTOOLCHAIN=go1.26.10 \
  go test -c -o /tmp/lifecycle.test ./pkg/lifecycle/
```

Copy binary plus files the tests read (`go.mod`, `role.yaml`, testdata):

```
SSH='ssh -o IdentityAgent=none -o IdentitiesOnly=yes \
  -o IdentityFile=$HOME/.ssh/opi_lab_key -o UseKeychain=yes \
  -o KexAlgorithms=curve25519-sha256'
SCP='scp -o IdentityAgent=none -o IdentitiesOnly=yes \
  -o IdentityFile=$HOME/.ssh/opi_lab_key -o UseKeychain=yes \
  -o KexAlgorithms=curve25519-sha256'

JOB=/tmp/lab-job-lcm-from-start
$SSH root@172.22.1.2 "rm -rf $JOB && mkdir -p $JOB/pkg/lifecycle/testdata $JOB/config/rbac"

$SCP /tmp/lifecycle.test root@172.22.1.2:$JOB/lifecycle.test
$SCP go.mod go.sum root@172.22.1.2:$JOB/
$SCP config/rbac/role.yaml root@172.22.1.2:$JOB/config/rbac/role.yaml
$SCP -r pkg/lifecycle/testdata/lcm-probe \
  root@172.22.1.2:$JOB/pkg/lifecycle/testdata/
$SSH root@172.22.1.2 "chmod +x $JOB/lifecycle.test"
```

From another Mac, wrap those `$SSH` / `$SCP` lines in:

```
ssh kartikey0104@Kartikeys-MacBook-Pro.local '...'
```

2. On dh2 — start: cluster and LCM object

```
hostname
k3s --version
k3s kubectl get nodes -o wide
k3s kubectl get dpuoperatorconfig -o yaml
k3s kubectl get dpuoperatorconfig dpu-operator-config \
  -o jsonpath='{range .status.conditions[*]}{.type}={.status}/{.reason}{"\n"}{end}'; echo
```

Expect: `spec.vendor=nvidia`, `adopt-existing=true`, `VendorOperatorSupported=True`, `VendorOperatorReady=True`. `VendorDPUReady` is False/Rebooting if the new observer image is rolled.

3. On dh2 — prereqs (must be present, do not install)

```
k3s kubectl get deploy -A | egrep 'cert-manager|argo|nfd|maintenance|dpf-operator|opi-nvidia'
k3s kubectl get crd | egrep 'dpuoperatorconfigs|dpus.provisioning|dpfoperatorconfigs'
k3s kubectl get ns cert-manager dpf-operator-system
```

4. On dh2 — run the compiled LCM tests

```
JOB=/tmp/lab-job-lcm-from-start
cd $JOB/pkg/lifecycle
# binary looks for ../../go.mod and ../../config/rbac/role.yaml
$JOB/lifecycle.test -test.count=1 -test.timeout=8m -test.v
echo EXIT:$?
```

Named slices if the full binary is too noisy:

```
$JOB/lifecycle.test -test.count=1 -test.timeout=8m \
  -test.run 'TestLifecycle|TestHelm|TestSecretHelm|TestClusterHelm|TestClaim|TestObserveNVIDIA|TestDPU|TestValidate
echo EXIT:$?
```

First fail mode we already hit: missing `go.mod` → `TestPatchedGovulnDependencies` FAIL. The copy in section 1 includes it.

5. On dh2 — Helm pin / claim (read-only)

```
k3s kubectl -n dpf-operator-system get secret \
  -l name=dpf-operator \
  -o jsonpath='{range .items[*]}{.metadata.name} owner={.metadata.labels.owner} mode={.metadata.labels.lifecycle\.o

k3s helm -n dpf-operator-system list 2>/dev/null || \
  k3s kubectl -n dpf-operator-system get secret -l owner=helm
```

Expect one Secret `sh.helm.release.v1.dpf-operator.v1`, storage `owner=helm`, adapter owner `opi-nvidia-dpf-adapter`, `mode=adopted`. Do **not** `helm uninstall` / `helm upgrade`.

6. On dh2 — RBAC (no wildcards)

```
k3s kubectl get clusterrole opi-nvidia-dpf-adapter-manager-role -o yaml \
  | grep -E '^\s+- \*' || echo 'NO_WILDCARDS'
k3s kubectl get clusterrole opi-nvidia-dpf-adapter-manager-role -o yaml \
  | egrep 'dpuoperatorconfigs|secrets|dpus|subscriptions'
k3s kubectl -n dpf-operator-system get deploy -l control-plane=controller-manager \
  -o jsonpath='{range .items[*]}{.metadata.name} {.spec.template.spec.containers[0].image}{"\n"}{end}'
```

Last roll: image `controller:lcm-20261024b`. Watch errors on Secrets must be 0:

```
POD=$(k3s kubectl -n dpf-operator-system get pod -l control-plane=controller-manager -o jsonpath='{.items[0].metada
```

```
k3s kubectl -n dpf-operator-system logs "$POD" --tail=200 | grep -c 'secrets is forbidden' || true
```

7. On dh2 — DPU / user-flow stall (observe only)

```
k3s kubectl -n dpf-operator-system get dpu dh2-bluefield-2 \
  -o jsonpath='phase={.status.phase} method={.status.rebootStatus.method} rebootPhase={.status.rebootStatus.phase}{
k3s kubectl -n dpf-operator-system get dpu dh2-bluefield-2 -o yaml \
  | egrep 'host-power-cycle-required|phase:|SystemLevelReset|NVCONFIG|PF_SF|PF_TOTAL'
k3s kubectl get dataprocessingunit -o wide
k3s kubectl -n dpf-operator-system get dpuservice
```

Expect DPU **Rebooting**, `host-power-cycle-required=true`. Dataplane Pending is the known stall. Do **not** run `mlxfwreset`. Do **not** reboot the host.

8. On dh2 — AMD check (must stay NotFound on BlueField)

```
k3s kubectl api-resources | grep -i dscdevice || echo 'dscdevices.dpu.amd.com NotFound (expected on BF2)'
```

Do **not** apply AMD CRD stubs.

9. Cleanup job files on dh2

```
rm -rf /tmp/lab-job-lcm-from-start
ls /tmp/lab-job-lcm-from-start 2>/dev/null || echo 'job dir gone'
```

Pass / fail you should see

Check	Pass
`lifecycle.test`	exit 0
Helm secret	one pin v26.4.0, claimed, not a second release
ClusterRole	no *
LCM conditions	Supported True, VendorOperatorReady True
DPU	Rebooting reported; no reset run
dscdevices	NotFound on this BlueField
job dir	deleted